

Primary VPT Chain: ProducerUA -> ProZorro -> Retail

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely I(1)-like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

standardized_type=sour_cream

retailer=silpo

promo_variant=promo_controlled

frequency=daily

Primary chain is strictly ProducerUA -> ProZorro -> Retail. Retail is estimated separately for silpo, novus, and silpo_novus. Combined rule: daily median of available silpo and novus standardized_type prices.

Table: PreTests

series	adf_level_p	kpss_level_p	adf_diff1_p	kpss_diff1_p	adf_diff2_p	kpss_diff2_p	integration_class	stability_flag	standardized_type	retailer	promo_variant	cointegration_p	pair_n_obs
producer	0.014081535169907735	0.01	4.706435460154828e-13	0.1	3.085032729165042e-21	0.1	I(1)	1.0	sour_cream	silpo	promo_controlled	nan	nan
prozorro	0.21880956753864628	0.04947047925942609	2.411890357322739e-12	0.1	3.969186107838014e-13	0.1	I(1)	0.0	sour_cream	silpo	promo_controlled	nan	nan
retail	0.0001866404404338293	0.1	1.2992938855568006e-06	0.1	1.670971302457964e-10	0.1	I(0)	0.0	sour_cream	silpo	promo_controlled	nan	nan
producer_to_prozorro	nan	nan	nan	nan	nan	nan	pair	nan	sour_cream	silpo	promo_controlled	0.3418935830778411	178.0
prozorro_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	sour_cream	silpo	promo_controlled	0.0009999496932936124	38.0
producer_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	sour_cream	silpo	promo_controlled	0.9914737446395873	48.0

Table: ModelCoefficients

standardized_type	retailer	promo_variant	frequency	link	model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	notes	primary_chain_doc
sour_cream	silpo	promo_controlled	daily	producer_to_prozorro	ARDL	prozorro	producer	178	0.1503321327484457	0.53201805897649	nan	nan	nan	nan	nan	ok	0	bounds_lower_p=0.1226; bo...	Primary chain is strictly...
sour_cream	silpo	promo_controlled	daily	producer_to_prozorro	NARDL	prozorro	producer	176	0.8049124174362647	0.3181803694069162	-0.9103202976511395	1.2936168519519301e-25	0.2798213093974368	0.3401073440796139	nan	unreliable	1	Wald short/long asymmetry...	Primary chain is strictly...
sour_cream	silpo	promo_controlled	daily	prozorro_to_retail	ARDL	retail	prozorro	38	-0.09602698663315934	-0.0780821130244814	nan	nan	nan	nan	nan	ok	0	bounds_lower_p=nan; bound...	Primary chain is strictly...
sour_cream	silpo	promo_controlled	daily	prozorro_to_retail	NARDL	retail	prozorro	36	-0.02591699786459659	-0.02222537492071102	-0.8330689213690337	0.00037765858299200743	0.7464656966292028	0.8267608548502566	nan	ok	0	Wald short/long asymmetry...	Primary chain is strictly...

Table: ResidualDiagnostics

model_family	link	y_series	x_series	ljungbox_p	arch_p	jb_p	unreliable_flag
ARDL	producer_to_prozorro	prozorro	producer	0.9991919902903724	0.4332665662567991	0.028152037118155036	0
NARDL	producer_to_prozorro	prozorro	producer	0.0725257366713705	0.012783066753899433	0.03261969684847455	1
ARDL	prozorro_to_retail	retail	prozorro	0.9744669047601868	1.0	1.858168228265059e-39	0
NARDL	prozorro_to_retail	retail	prozorro	0.7865388890305411	0.3344492053876417	0.29874884459635026	0

Table: SeriesUsed

date	producer	prozorro	retail	standardized_type	retailer	promo_variant	combined_rule
2021-02-01	12.7168318078645	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-02	12.72632554286918	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-03	12.73569490777093	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-04	12.74493705667296	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-05	12.75404914367847	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-06	12.76302832289066	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-07	12.77187174841275	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-08	12.78057657434793	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-09	12.78861583935331	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-10	12.79582084628455	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-11	12.80272614554264	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-12	12.80986628752861	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-13	12.81777582264346	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-14	12.8269893012882	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-15	12.83804127386385	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-16	12.85602638141248	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-17	12.88286036370906	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-18	12.91430941925396	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-19	12.94613974654752	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-20	12.97411754409009	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-21	12.99400901038205	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-22	13.00158034392373	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-23	12.85209968138777	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-24	12.46659692011084	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-25	11.93948089958932	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-26	11.3651604593196	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-27	10.83804443879808	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-28	10.45254167752114	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-03-01	10.30306101498519	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-03-02	10.3211976511164	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-03-03	10.367971081139	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-03-04	10.43192658749644	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-03-05	10.50160945263215	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-03-06	10.56556495898959	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-03-07	10.61233838901219	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-03-08	10.6304750251434	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-03-09	10.62692996199335	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-03-10	10.6172401963284	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-03-11	10.60282386382634	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-03-12	10.58509910016495	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...

Table: LagProfile

	pair	lag	corr
producer_to_prozorro	1	0.07433869937437788	
producer_to_prozorro	2	0.06992193974139833	
producer_to_prozorro	3	0.05247780353970976	
producer_to_prozorro	4	0.03707093523380621	
producer_to_prozorro	5	0.04488476174629457	
producer_to_prozorro	6	0.03942159255359992	
producer_to_prozorro	7	0.04774350069192749	
producer_to_prozorro	8	0.055533922878419725	
producer_to_prozorro	9	0.05815494362517447	
producer_to_prozorro	10	0.05835686030753	
producer_to_prozorro	11	0.05572808262664563	
producer_to_prozorro	12	0.05132684090763728	
producer_to_prozorro	13	0.0478799443885565	
producer_to_prozorro	14	0.0459078796867533	
producer_to_prozorro	15	0.04627227965570159	
producer_to_prozorro	16	0.04980743752852967	
producer_to_prozorro	17	0.05457381656925348	
producer_to_prozorro	18	0.05403494026121064	
producer_to_prozorro	19	0.045342075346488105	
producer_to_prozorro	20	0.030030963601459588	
producer_to_prozorro	21	0.011140301716297355	
producer_to_prozorro	22	-0.006115614807159968	
producer_to_prozorro	23	-0.017626157690186797	
producer_to_prozorro	24	-0.022831222527256615	
producer_to_prozorro	25	-0.024681880076761423	
producer_to_prozorro	26	-0.028172449275670572	
producer_to_prozorro	27	-0.03484368926283701	
producer_to_prozorro	28	-0.04221286356527852	
producer_to_prozorro	29	-0.04827032698871696	
producer_to_prozorro	30	-0.05291303035138733	
prozorro_to_retail	1	-0.03667757521279544	
prozorro_to_retail	2	-0.27789900612579105	
prozorro_to_retail	3	-0.22461696407750467	
prozorro_to_retail	4	-0.18260700540033242	
prozorro_to_retail	5	-0.019443231840724415	
prozorro_to_retail	6	-0.09530230617623504	
prozorro_to_retail	7	0.11332668653099863	
prozorro_to_retail	8	-0.13332162162664174	
prozorro_to_retail	9	-0.024746620429041553	
prozorro_to_retail	10	0.09934959399577038	

Table: ModelEligibility

standardized_type	retailer	promo_variant	frequency	integration_producer	integration_prozorro	integration_retail	n_obs_producer	n_obs_prozorro	n_obs_retail	n_obs_pair_prod_prozorro	n_obs_pair_prozorro_retail	any_i2	eligibility_note
sour_cream	silpo	promo_controlled	daily	I(1)	I(1)	I(0)	1793	188	48	178	38	0	Admissible for ARDL/ECM s...

Table: NARDL_Multipliers

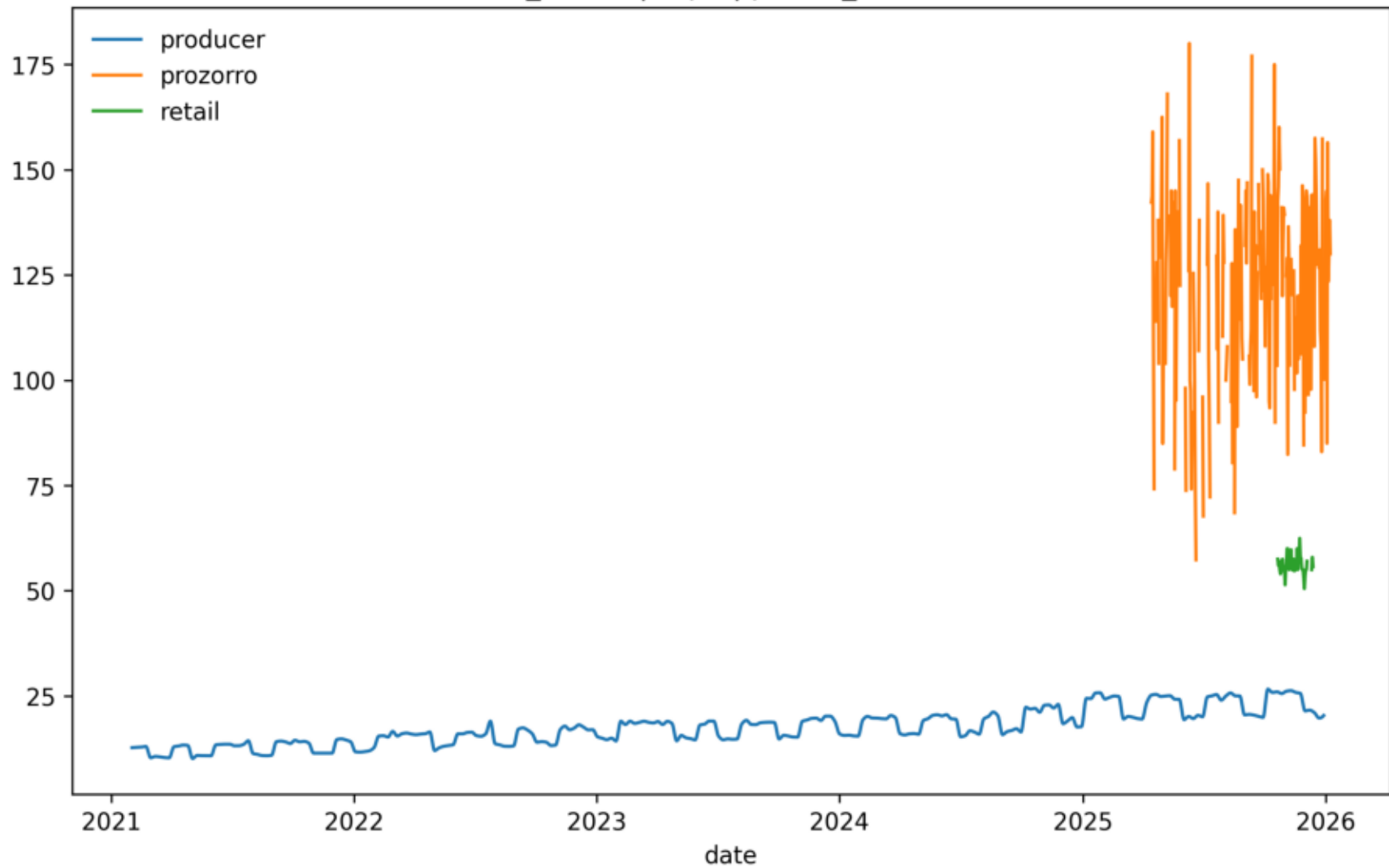
horizon	mult_pos	mult_neg	link
0	0.5272629908128263	-0.27764942662343833	producer_to_prozorro
1	0.5660151486488385	-0.29805577903403124	producer_to_prozorro
2	0.5688633093639157	-0.2995555812262039	producer_to_prozorro
3	0.5690726401447521	-0.2996658119313484	producer_to_prozorro
4	0.5690880252933407	-0.29967391353862705	producer_to_prozorro
5	0.5690891560529745	-0.29967450898107123	producer_to_prozorro
6	0.5690892391602219	-0.2996745527442027	producer_to_prozorro
7	0.569089245268341	-0.29967455596065407	producer_to_prozorro
8	0.5690892457172685	-0.2996745561970531	producer_to_prozorro
9	0.5690892457502632	-0.2996745562144276	producer_to_prozorro
10	0.5690892457526882	-0.2996745562157046	producer_to_prozorro
11	0.5690892457528665	-0.29967455621579847	producer_to_prozorro
12	0.5690892457528796	-0.29967455621580535	producer_to_prozorro
13	0.5690892457528804	-0.29967455621580585	producer_to_prozorro
14	0.5690892457528806	-0.2996745562158059	producer_to_prozorro
15	0.5690892457528806	-0.2996745562158059	producer_to_prozorro
16	0.5690892457528806	-0.2996745562158059	producer_to_prozorro
17	0.5690892457528806	-0.2996745562158059	producer_to_prozorro
18	0.5690892457528806	-0.2996745562158059	producer_to_prozorro
19	0.5690892457528806	-0.2996745562158059	producer_to_prozorro
20	0.5690892457528806	-0.2996745562158059	producer_to_prozorro
0	0.0031402000259013324	0.029057197890497922	prozorro_to_retail
1	0.003068300511629963	0.028391890458747873	prozorro_to_retail
2	0.0030699467571270074	0.028407123654337132	prozorro_to_retail
3	0.003069909063906629	0.028406774867841166	prozorro_to_retail
4	0.003069909926948542	0.02840678285382277	prozorro_to_retail
5	0.0030699099071879233	0.028406782670971936	prozorro_to_retail
6	0.0030699099076403717	0.028406782675158577	prozorro_to_retail
7	0.0030699099076300124	0.028406782675062716	prozorro_to_retail
8	0.0030699099076302496	0.028406782675064912	prozorro_to_retail
9	0.003069909907630244	0.028406782675064864	prozorro_to_retail
10	0.0030699099076302444	0.028406782675064864	prozorro_to_retail
11	0.0030699099076302444	0.028406782675064864	prozorro_to_retail
12	0.0030699099076302444	0.028406782675064864	prozorro_to_retail
13	0.0030699099076302444	0.028406782675064864	prozorro_to_retail
14	0.0030699099076302444	0.028406782675064864	prozorro_to_retail
15	0.0030699099076302444	0.028406782675064864	prozorro_to_retail
16	0.0030699099076302444	0.028406782675064864	prozorro_to_retail
17	0.0030699099076302444	0.028406782675064864	prozorro_to_retail
18	0.0030699099076302444	0.028406782675064864	prozorro_to_retail

Table: VECM_IRF

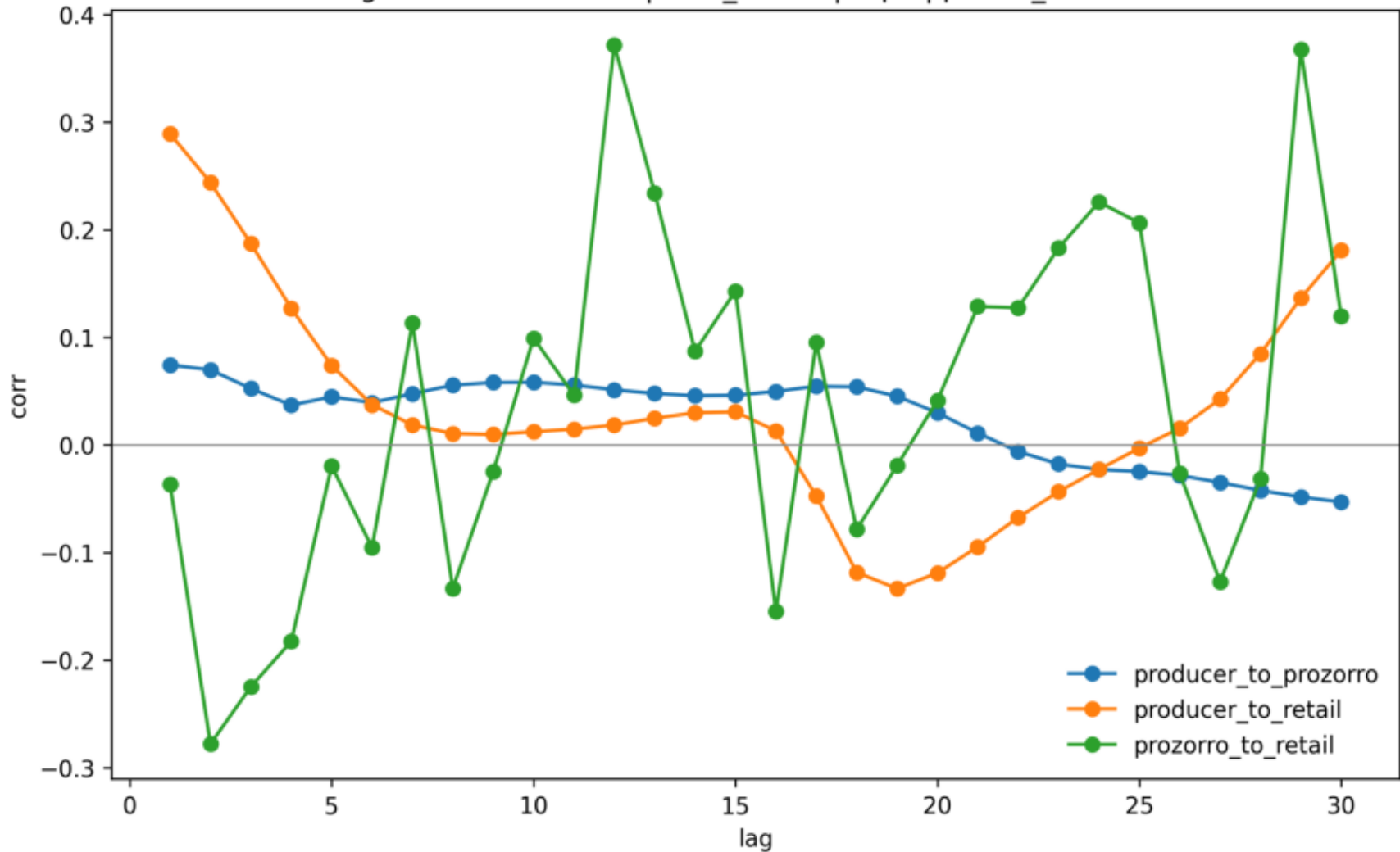
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time_series_promo_controlled.png

sour_cream | silpo | promo_controlled



Lag-Correlation Profile | sour_cream | silpo | promo_controlled



NARDL Dynamic Multipliers | sour_cream | silpo | promo_controlled

